Highlights

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Abstract

This is the abstract.

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1. Introduction

The mountainous central European country of Switzerland, like the majority of countries globally, exhibits declining biodiversity due to multiple drivers such as urbanization (Concepción et al., 2016), climatic change(Vittoz et al., 2013) and environmentally deleterious agricultural practices (Chaudhary et al., 2016). To address biodiversity loss, in 2012, the federal government of Switzerland ratified the Swiss Biodiversity Strategy (SBS)(FOEN, 2012). One of the goals of the SBS is the development of Switzerland's Ecological Infrastructure (EI) as a "network of protected and connected areas that contribute to safeguarding the essential services of ecosystems for society and economy … and form the basis for a rich and resilient biodiversity" (FOEN, 2023). Achieving this goal will require consideration of not just how Switzerland's EI should be planned under existing conditions but how it's development in space and time could proceed in concurrence with changes in the boundary conditions of the socio-ecological system within which it is situated.

In this regard, Mayer et al. (2023) defined 5 narrative scenarios to frame the development of EI in Switzerland between 2020 and 2060 alongside expected demographic, economic and climatic changes in Switzerland. The scenarios were produced through a participatory process with the aim of envisioning nature-positive outcomes using the IPBES Nature futures framework (Pereira et al., 2020; Durán et al., 2023). As such the scenarios can characterized as both exploratory and normative, encapsulating the development of EI from different perspectives of societal-nature relationships (Mayer et al., 2023). These perspectives are reflected in the names of the scenarios which are as follows: EI for Nature (EI-NAT); EI for Society (EI-SOC); EI as Culture (EI-CUL); Business-as-Usual (BAU); and Growth and Extinction (GR-EX).

However, whilst descriptive scenario narratives are a useful starting point for planners and decision-makers, a valuable next step is to operationalise the narratives through computer-based simulation modelling so that their spatio-temporal impacts on critical aspects of the study system can be visualized and explored in detail (Verburg et al., 2004, 2006). A key benefit of such modelling is that it can be used to confirm the coherency of scenarios (Volkery et al., 2008) as well as to elucidate the balance between their perceived desirability and plausibility thereby increasing legitimacy in the eyes of stakeholders(Iwaniec et al., 2020). This is of particular importance for normative scenarios which strive for explicitly desirable outcomes (Nassauer and Corry, 2004) that often necessitate radical or transformative system changes and hence may be considered to lack plausibilty (Wiek and Iwaniec, 2014). Simulation modelling is also important when scenarios would likely result in trade-offs that must be negotiated by society (Rounsevell et al., 2012). This is pertinent in the case of EI in Switzerland, which, under the definition employed by the SBS, aims to balance both ecosystem service (ES) provision and biodiversity protection, which have often been acknowledged as discordant objectives (Cordingley et al., 2016).

In terms of operationalizing the specific scenarios of EI development from

Mayer et al. (2023), the proximal linkages between EI as a "network of areas" at the landscape level in Switzerland and it's intended outcomes for biodiversity and ES are difficult to model directly. Instead, it is logical to focus on a higher level of abstraction and model an intermediary, or distal, interaction between the physical expression of EI and these outcomes. A common component in the modelling of both measures of biodiversity (Gerecke et al., 2019) and ES provision (Burkhard et al., 2009; Newbold et al., 2015; Gomes et al., 2021) is land use and land cover (LULC). Furthermore LULC change (LULCC) trends are also a central theme in the SBS concept of EI as well as the scenarios of Mayer et al. (2023). Specifically, LULCC related trends such as urban sprawl, the spatial distribution of agricultural production and the expansion of protected areas to meet international targets, feature heavily in the narratives (ibid.). On this basis, the goal of this research is to operationalize the scenario's of EI development produced for Switzerland by Mayer et al. (2023) by simulating their implications for trends in LULCC in Switzerland between 2020 and 2060 using a temporally dynamic Constrained Cellular Automata (CCA) model.

CCA models have enjoyed enduring popularity for the modelling of scenarios of LULCC for the last three decades (Triantakonstantis and Mountrakis, 2012; Aburas et al., 2016). Thompson et al. (2020) ascribe this popularity to the fact that CCAs represent LULCC as a phenomenon rather than as a process, i,e they abstract the actions of the actors involved, which can make them easier to explain to non-experts. LULCC-CCAs treat landscapes as a pixelated grid of cells of individual LULC states with cells changing state over time typically through the combination of two mechanisms. Firstly, the spatial distribution of potential LULCC is determined through calculation of cellular probabilities of change, often referred to as transition potential (TP), to alternate LULC states (White and Engelen, 1997). The method by which this is achieved, differs between popular LULCC-CAs, although it is commonly based on the quantification of correlative relationships with explanatory variables such as biophysical, socio-economic and neighbourhood (focal LULC) predictors using a statistical model (Tobler, 1979; García-Álvarez, 2018; Ren et al., 2019). Secondly, the amount of cellular LULC changes that occur in a given simulation time step is prescribed by values of transition rates extrapolated from historic LULC data, with a model specific allocation algorithm selecting the required number of cells to change based upon their transition probabilities (Mas et al., 2018).

The two general mechanisms of LULCC-CCAs offers several opportunities to operationalize aspects of scenarios. Firstly, the data supplied to predict cellular TP values can be both specific to the different scenario's being modelled as well as dynamic over the simulation time steps (Liao et al., 2023). For example, in scenarios contrasting differential levels of climatic change, the data provided may come from exogenous climatic models of different Relative Concentration Pathways (Song et al., 2019). A second method is to alter predicted cellular TP values according to proscribed criteria, this can be done directly: such as spatially designated land management policies, for example reducing the potential for transition to urban LULC states inside environmental protection

areas(Poelmans and Van Rompaey, 2010; Gharbia et al., 2016; Price et al., 2017; Clerici et al., 2019; Daneshi et al., 2021; Peng et al., 2021; Frank et al., 2023). Or indirectly by altering the influence of variables within the statistical model used to calculate TPs(Troupin and Carmel, 2016; Gago-Silva et al., 2017). Another means is to adjust parameters of the LULCC-CCA's allocation algorithm, that may for example increase the creation of new patches of LULC class cover relative to the expansion of existing areas (Prieto-Amparán et al., 2019). Finally, scenarios can be instantiated through individual specification of the transitions rates determining the quantities of LULCC (Shrestha et al., 2018; Cheng et al., 2020; Schirpke et al., 2020; Peng et al., 2021; Kiziridis et al., 2023; Molinero-Parejo et al., 2023). The influence of these mechanisms within a generalized schematic of an LULCC-CCA model are highlighted visually in Figure 1 below.

Many LULCC-CCA scenario simulation studies utilize one or several of these mechanisms, however it is rare that they are all implemented substantively in combination with each other. Possible reasons for this could be a lack of future projected data for LULCC drivers or that the scenarios being modelled offer only limited descriptions of expected trends in LULCC. Given the level of detail provided by the EI development scenarios (Mayer et al., 2023) as well as the fact that Switzerland represents a data-rich case study, this research aims to produce the most comprehensive simulations of LULCC between 2020 and 2060, for the country to date, incorporating:

i. Region specific modelling of >20 LULC class-class transitions using a tuned Random Forests approach
ii. Scenario specific transition rates informed by a participatory process
iii. Multiple variations of future climatic, economic, and demographic data
iv. The expansion of nature conservation areas to simulate the effect of meeting international targets, delineated using state of the art conservation prioritization maps
v. Expert feedback on simulation results.

2. Methods

2.1. Model overview

To perform scenario-based simulation of future LULCC in Switzerland between 2020 and 2060 we created a constrained cellular automata (CCA) model combining the statistical programming software R (4.2.2)(R core team, 2022) with the environment modelling software Dinamica EGO (7.2.0)(Soares-Filho et al., 2009). In the interest of open science the CCA model along with the research data and results has been made available alongside this manuscript (URL)

Figure 1 highlights the primary processes taking place within the LULCC-CCA model, the details of each these processes are summarized in the subsequent sections although for brevity some material has been presented as appendices.

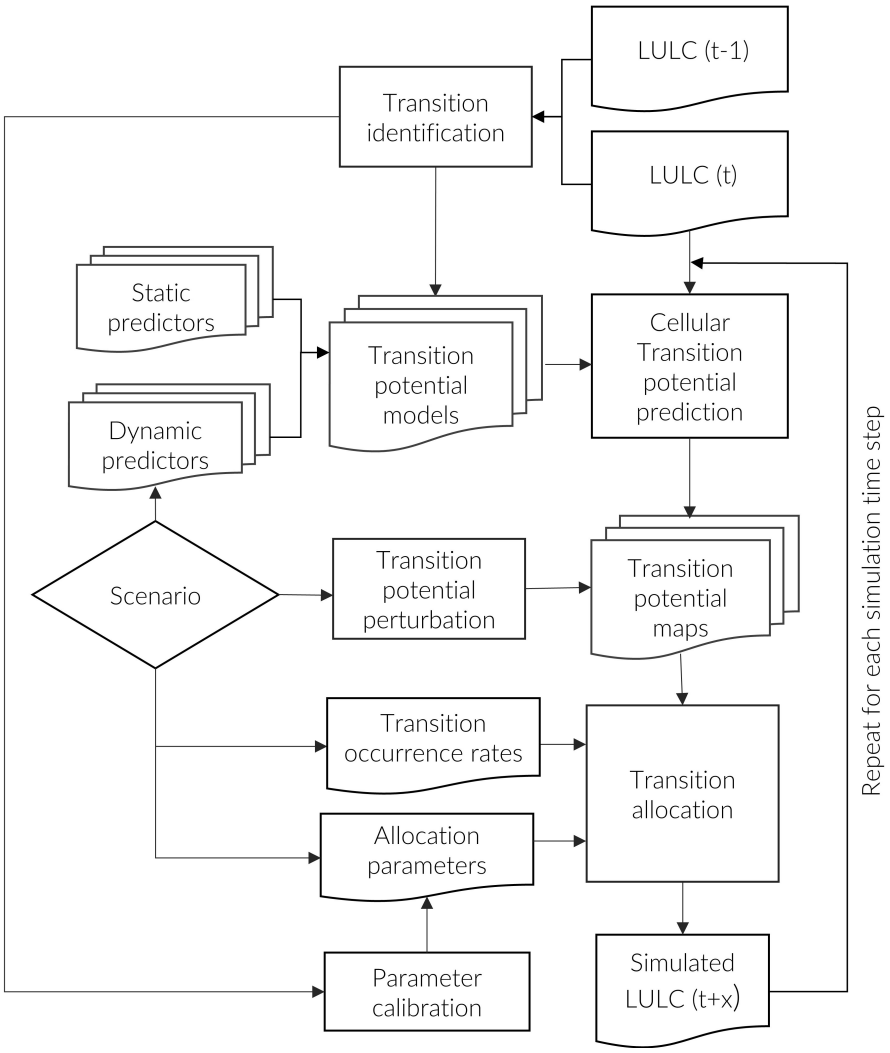

Figure 1: Flowchart showing the primary processes and inputs in the LULCC-CCA model

2.2. Land use land cover transition identification

The process of identifying LULC transitions, i.e. pixel changes from an initial specific LULC class to a different final LULC class, was based upon a preceding study by Black et al. (2023). We utilized classified LULC datasets for four time periods from the Swiss area statistics (SFSO, 2021): 1981-1985; 1992-1997; 2004-2009; 2013-2018 (Note that date range is the extent of the data-collection flying period, and in fact each product represents a single LULC map). The LULC data is originally presented as point data generated from aerial photography, however for the purpose of modelling all maps were converted to 100m resolution rasters. Given that transitions are identified between two time points having four historic LULC maps meant that we had three LULC transition periods, with which to calibrate our model, which will henceforth be referred to using the latter year of the flying period as: 1985-1997; 1997-2009; and 2009-2018 respectively. Following Black et al. (2023) we aggregated the original 72 LULC classes in the data to 10 classes: Alpine pastures; Closed forest; Glacier; Grassland; Intensive agriculture; Open Forest; Shrubland; Permanent crops; Settlement/urban/amenities; Static (see Appendix A for a table of class aggregations).

Instances of LULC transitions were identified by overlaying the maps for each transition period and extracting pixels on the basis of their specific changes in LULC classes between their initial class (start map) and final class (end map). With 10 aggregated classes this gives the potential for 100 possible class-class transitions which is unfeasible for modelling. Thus, as per Black et al. (2023), we filtered to viable LULC transitions for each period by first excluding illogical transitions (e.g from static to other classes) as well as applying a threshold to remove transitions that led to an a change in less than 0.5% of the total area of the initial class.

The instances of LULC transitions were then combined with instances of non-transitions (i.e. pixels with the same initial class but a different final class), coded as 1's and 0's respectively, to give seperate datasets for each transition. These transition datasets were first identified using the entire LULC map region (i.e the extent of Switzerland). However, to better reflect the potential for localized differences in LULCC processes the datasets were sub divided according to the six biogeographical regions of Switzerland (Gonseth et al., 2001). A table listing the transitions identified for the calibration periods is included as Appendix B .

2.3. Transition potential model calibration

The process of creating the statistical models to be used to iteratively predict cellular TP within the LULCC-CCA was refined from that of Black et al. (2023). The predictors chosen for the modelling were grouped into categories related to land suitability and accessibility as well as neighbourhood predictors capturing the influence of the LULC class of the surrounding cells. The details of the suitability and accessibility predictors are provided in Table 1 below.

Table 1: Predictor data used in statistical models of transition potential.

Variable name	Data source/s	Temporality
Distance to roads	Swisstopo, 2011	Static
Continentality	Descombes et al. (2020)	Static
Light	Descombes et al. (2020)	Static
Soil pH	Descombes et al. (2020)	Static
Soil nutrients	Descombes et al. (2020)	Static
Soil moisture	Descombes et al. (2020)	Static
Soil moisture variability	Descombes et al. (2020)	Static
Soil aeration	Descombes et al. (2020)	Static
Soil humus	Descombes et al. (2020)	Static
Mean elevation	Wiederkehr and Möri (2013)	Static
Aspect	Wiederkehr and Möri (2013)	Static
Slope	Wiederkehr and Möri (2013)	Static
Hillshade	Wiederkehr and Möri (2013)	Static
Noise pollution index	FOEN, 2009	Static
Distance to lakes (mean minimum dist to all lakes of different categories)	Swisstopo, 2022	Static
Distance to river (mean minimum distance (agg. From 25m data to rivers of all Strahler classes)	Swisstopo, 2007	Static
Average annual mean air temperature	Broennimann (2018)	Dynamic
Average annual precipitation	Broennimann (2018)	Dynamic
Average growing degree days heat sum above 0°C	Broennimann (2018)	Dynamic
Average growing degree days heat sum above 3°C	Broennimann (2018)	Dynamic
Average growing degree days heat sum above 5°C	Broennimann (2018)	Dynamic
Population per municipality	SFSO, 2016; SFSO, 2022a	Dynamic
Average annual change in no. of Full Time Equivalent (FTE) employees in the primary sector per labour market region	SFSO, 2013; SFSO, 2014a; SFSO, 2022b	Dynamic
Average annual change in no. of FTE employees in the secondary sector per labour market region	SFSO, 2013; SFSO, 2014b; SFSO, 2022b	Dynamic
Average annual change in no. of FTE employees in the tertiary sector per labour market region	SFSO, 2013; SFSO, 2014b; SFSO, 2022b	Dynamic

- approach refined from previous paper
- predictor filtering applied
- seperate models fitted for each calibration period

The accuracy of the statistical models used for transition potential prediction was assesed independently of the CCA model, using hold-out validation with multiple performance metrics, in a process that Black et al. (2023) refer to as stage 1 validation. These models are used for prediction

- The exception is glacier transition

2.4. Cellular automata parameter calibration

Dinamica EGO uses a combination of two algorithms for allocating cellular LULC transitions, Patcher and Expander, which represent the processes of the creation of new patches of LULC classes and the expansion of existing patches respectively(Soares-Filho et al., 2009). These algorithms require the calibration of 5 parameters for every transition: Mean patch size, Patch size variance, Patch isometry, and the relative % of transitions to be allocated by each algorithm (i.e % by patcher, % by expander). To calibrate these parameters we first used the historic LULC maps from the calibration periods to estimate parameter values using the R package SDMtools , which were then refined using brute force simulation. This involved applying a random permutation (from normal distribution with mean = 0 and sd = 0.05 effectively 5% bounding) to each transition parameter value for 100 repetitions to produce candidate sets of parameters values. These candidate sets were then tested by using them in the Dinamica EGO model to simulate LULCC between 2010 and 2020 (i.e covering the final calibration period) using a 5 year time step. The results of these simulations were used to identify the candidate set of parameter values that displayed the greatest value of the minimum similarity score produced by Dinamica EGO's "fuzzy reciprocal similarity" map comparison method (Soares-Filho et al., 2009, 2013). This set of parameter values was used as the basis for all future simulations with some scenarios involving deliberate modifications of values (2.5.2).

2.5. Scenario operationalisation

2.5.1. Transition rates

The cellular transition allocation process in Dinamica EGO is constrained by input in the form of rates of transitions (the % of the initial class area to be converted) for each time step in the simulation. To establish transition rates for each of our 5 future scenario's we iterated between quantitative estimation and expert elicitation.

Initially, transition rates were calculated for each of the calibration time periods before being extrapolated (linear models) over 5 year time steps between 2020 and 2060. These extrapolated transitions rates were dubbed the reference rates and were applied to the values of LULC class extents in the final calibration year (2018) to estimate reference class areas for 2060.

These reference class areas were used as a prompt in discussions with experts involved in the scenario creation process to code whether statements within the scenario's were perceived as likely to lead to an increase or decrease in the areas of certain LULC classes relative to the reference areas. Experts were encouraged to focus only on the direct impacts of statements and not potential indirect impacts, i.e statements referencing change in agriculture practices were linked to agricultural LULC classes only and the potential for subsequent impacts on forestry were only considered in corresponding statements. Importantly, experts were able to allocate different magnitudes of impact on LULC classes due to individual statements. This was realized in a simplistic manner through the allocation of a desired number of either pluses and minuses, each representing an equally weighted percentage change to the class area, which were summed for each class across all statements to give overall percentage impacts of the scenario on class areas. This process was first employed for the BAU scenario with the percentage class impacts used to modify the reference 2060 class areas to give projected 2060 class areas. THis was then repeated for the remaining scenarios (EI-NAT, EI-CUL, EI-SOC and GR-EX) however their areas were derived relative to the BAU rather than the reference areas. This is an important distinction as the scenarios are intended to represent deliberate changes to the status-quo represented by the BAU narrative.

Finally, it was necessary to dis-aggregate the projected class areas for each scenario back to areas and then rates for each transition. Given that it is possible to either increase or decrease the rates of transitions that result in the expansion of LULC class area simultaneous to altering transitions that result in class area contraction, this process of dis-aggregation represented an optimization problem. Our method to address this was to perform a first pass over the LULC classes and allocate areas to relevant transitions with respect to the proportion of area expansion vs. contraction exhibited by the LULC class in the calibration periods. In the case that there were multiple transitions contributing to either class expansion or contraction then the values were ascribed based on the relative contributions of each transition to historic class change rates. This gave us an approximation of the transition areal changes for each scenario across all time steps with further manual changes made in order to reduce the discrepancy between the sum of these and the expert projected class areas for 2060.

2.5.2. Manipulation of transition potential and allocation parameters

From the scenarios details (Mayer et al., 2023) we extracted statements, grouped according to drivers, that pertain to spatialized differences in the locations LULCC processes . From these statements we derived 8 corresponding spatial intervention mechanisms which were applied across the scenarios to give a total of 19 interventions. The majority of the interventions functioned by directly alterating cellular transition probabilities during the simulation time. , however one intervention instead involved alteration of the Dinamica EGO allocation parameters. The details of these spatial interventions are lengthy and as such are presented in Appendix C.

2.5.3. Preparation of dynamic predictors

For future data for the predictors of average increase in full time equivalents per labour market region we utilised the future economic sector projections produced by Cretegny and Müller (2020).

Table 2: Variants of dynamic predictor data utilised for LULCC scenarios.

LULCC Scenario	Population growth scenario(SFSO, 2020)	Economic scenario(Cretegny and Müller, 2020)	Climate change scenarios(Broennimann, 2018)
BAU	Reference	Reference - Central	RCP45
EI-NAT	Low	Ecological awareness - Urban densification	RCP26
EI-CUL	Reference	Ecological awareness-Central	RCP26
EI-SOC	Reference	Combined technological accelaration and Ecological awareness - Urban densification	RCP45
GR-EX	High	Reference - Peri-Urbanisation	RCP85

2.6. Simulation and validation

reference Van Vliet's 2016 review re: validation

3. Results

3.1. Scenario simulation maps

3.2. Landscape patterns

3.3. Expert validation

- Maps of all time points between 2020-2060 included with research data

4. Discussion

- Difficulty in deriving rates of transitions for multiple scenarios (Karner et al., 2019)
- For cellular lulcc modelling non-stationarity/uncertainty are key limitations
- choice of regions
- still requires stationary data for some sources i.e. roads.

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Appendix A. Details of land use and land cover aggregation

Table A1: Aggregation of the Swiss area statistics land use land cover (LULC) classes.

Aggregated LULC class	Area statistics LULC classes
Alpine pastures	Favorable alpine pastures, Brush alpine pastures, Rocky alpine pastures, Sheep pastures
Closed forest	Normal dense forest, Forest strips, Afforestations, Felling areas, Brush forest
Glacier	Glaciers, perpetual snow
Grassland or meadows	Meadows, Farm pastures, Brush meadows and farm pastures, Alpine meadows
Intensive agriculture	Arable land
Open Forest	Damaged forest areas, Open forest (on agricultural areas), Open forest (on unproductive areas), Groves, hedges, Clusters of trees (on agricultural areas), Clusters of trees (on unproductive areas)
Overgrown vegetation	Scrub vegetation, Unproductive grass und shrubs
Permanent crops	Intensive orchards, Field fruit trees, Vineyards, Horticulture
Settlement/urban/industrial/and cities	Industrial and commercial buildings, Surroundings of industrial and commercial buildings, One- and two-family houses, Surroundings of one- and two-family houses, Terraced houses, Surroundings of terraced houses, Blocks of flats, Surroundings of blocks of flats, Public buildings, Surroundings of public buildings, Agricultural buildings, Surroundings of agricultural buildings, Unspecified buildings, Surroundings of unspecified buildings, Parking areas, Construction sites, Unexploited urban areas, Public parks, Sports facilities, Golf courses, Camping areas, Garden allotments, Cemeteries

Aggregated LULC class	Area statistics LULC classes
Static class	Motorways, Green motorway environs, Roads and paths, Green road environs, Sealed railway areas, Green railway environs, Airports, Airfields, green airport environs, Energy supply plants, Waste water treatment plants, Other supply or waste treatment plants, Dumps, Quarries, mines, Lakes, Rivers, Flood protection structures, Avalanche and rockfall barriers, Wetlands, Alpine sports facilities, Rocks, Screes, sand, Landscape interventions

Appendix B. Land use and land cover transitions identified

Appendix C. Details of manipulation of transition potential and allocation parameters

Table A2: Details of interventions to spatially manipulate cellular transition proabilities

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-SOC	"Land management is spatially zoned: areas for agriculture, residence, recreation, biodiversity promotion are spatially strongly separated."	-	Urban; Int_Ag; Open_Forest; Grassland; Perm_Crops, Alp_Past	spatial zoning	alter values of allocation parameter (patcher vs. Expander)	Alter values of allocation parameters : Increase % of transitions with expander (incurring decrease in patcher %)

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-NAT	"strong limitation of settlement growth and land fragmentation"	-	Urban; Static	spatial zoning	alter values of allocation parameter (patcher vs. Expander)	Alter values of allocation parameters : Increase % of transitions with expander (incurring decrease in patcher %)

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-SOC	"PAs are expanded and newly established (17% of Swiss land area protected by 2030, 22% by 2060); Site selection for new PAs prioritizing ecosystems important for NCP provision; site-specific management under IUCN categories II or IV"	Spatial prioritization map produced with the conservation prioritization software Zonation (Lehtomäki and Moilanen, 2013) using modelled spatial data for 7 NCPs: nutrient retention; recreation; landscape suitability for pollination; pest control; hazard regulation; carbon storage in vegetation; air quality regulation.	Urban, Int_AG, Perm_crops	Protection	Reduction of probability of transition to non-natural land classes inside PAs	Reduction of probability of transition to Urban, Int_AG or Perm_crops to 0 inside PAs

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-CUL	"PAs are expanded and newly established (17% of Swiss land area protected by 2030, 25% by 2060);Site selection for new PAs prioritizing species and landscape elements important for local communities and cultural heritage; Site-specific management under IUCN categories V or VI"	NA	Urban, Int_AG, Perm_crops	Protection	Reduction of probability of transition to non-natural land classes inside PAs	If the % difference between the means of the 90th percentile data points inside vs. outside the protected areas is negative then decrease the probabilities to the target classes of the data points inside the protected by the % difference value otherwise if it is positive then increase the probabilities to the target classes of the data points outside the protected areas by the % difference value.

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-NAT	"PAs are expanded and newly established (20% of Swiss land area protected by 2030, 30% by 2060); Site selection for new PAs prioritizing species and habitats important for biodiversity, ecosystem processes and functions; site-specific management under IUCN category Ia or IV"	NA	Urban, Int_AG, Perm_crops	Protection	Reduction of probability of transition to non-natural land classes inside PAs	Reduction of probability of transition to Urban, Int_AG or Perm_crops to 0 inside PAs

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-CUL	"No further urban sprawl"	NA	Urban	Urban densification	probability of Urban transitions increased inside building zones (using all classes in the building zones)	Increase probability to urban of cells inside building zone to 100%
EI-NAT	"No further urban sprawl."	NA	Urban	Urban densification	probability of Urban transitions increased inside building zones (using all classes in the building zones)	Increase probability to urban of cells inside building zone to 100%
EI-SOC	"No further urban sprawl."	NA	Urban	Urban densification	probability of Urban transitions increased inside building zones (using all classes in the building zones)	Increase probability to urban of cells inside building zone to 100%

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
GR-EX	"Increase in buildings outside the construction zone;Industrial areas develop in an uncoordinated manner."	NA	Urban	Urban sprawl	Minor increase in probability of cells to urban outside the building zones	If the % difference between the means of the 90th percentile data points outside vs. inside the building zones is positive then increase the probabilities to urban of the data points outside the building zones by the % difference value otherwise if it is negative then decrease the probabilities to urban of the data points inside the building zones by the % difference value.

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-NAT	"Development of small towns in mountain areas with spatial restrictions for settlement development."	NA	Urban	Mountain development	Use Typology of municipalities with 25 categories to identify cells and increase probability to urban	If the % difference between the means of the 90th percentile data points outside vs. inside the mountainous municipalities is positive then increase the probability of instances above the 90th percentile for the pixels inside the mountainous municipalities by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile outside the

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-SOC	"Development of small towns in mountain areas with spatial restrictions for settlement development."	NA	Urban	Mountain development	Use Typology of municipalities with 25 categories to identify cells and increase probability to urban	If the % difference between the means of the 90th percentile data points outside vs. inside the mountainous municipalities is positive then increase the probability of instances above the 90th percentile for the pixels inside the mountainous municipalities by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile outside the

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
BAU	"Abandonment of settlements in remote rural areas."	NA	Urban	Urban migration	Use Typology of municipalities with 25 categories to identify cells and decrease probability to urban	If the % difference between the means of the 90th percentile data points outside vs. the 80th percentile data points inside the remote rural municipalities is positive then increase the probability of instances above the 90th percentile for the outside pixels by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile inside the

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-NAT	"Abandonment of settlements in remote rural areas."	NA	Urban	Urban migration	Use Typology of municipalities with 25 categories to identify cells and decrease probability to urban	If the % difference between the means of the 90th percentile data points outside vs. the 80th percentile data points inside the remote rural municipalities is positive then increase the probability of instances above the 90th percentile for the outside pixels by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile inside the

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-SOC	"Abandonment of settlements in remote rural areas."	NA	Urban	Urban migration	Use Typology of municipalities with 25 categories to identify cells and decrease probability to urban	If the % difference between the means of the 90th percentile data points outside vs. the 80th percentile data points inside the remote rural municipalities is positive then increase the probability of instances above the 90th percentile for the outside pixels by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile inside the

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-CUL	"Revival of settlements in remote rural areas"	NA	Urban	Rural migration	Use Typology of municipalities with 25 categories to identify cells and increase probability to urban	If the % difference between the means of the 90th percentile data points outside vs. inside the remote rural municipalities is positive then increase the probability of instances above the 90th percentile for the pixels inside the remote rural municipalities by the % difference value. Else if the % difference is negative then decrease the probability of the instances above the 90th percentile outside

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-SOC	"Abandonment of poorly accessible marginal land"	NA	Alp_Past	Agri abandonment	Increase in probability of transition of marginal agricultural land to non-agricultural classes	Identify marginal agricultural land using an index (0-1) of slope, distance to roads, inverse distance to building zones and elevation (Müller 2022). Select upper quartile of most marginal pixels and then select 90th percentile of pixels according to probability and increase probability to transition from Agriculture to non-agricultural classes. Simultaneously select pixels below upper quartile of

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
BAU	"Abandonment of poorly accessible marginal land"	NA	Alp_Past	Agri abandonment	Increase in probability of transition of marginal agricultural land to non-agricultural classes	Identify marginal agricultural land using an index (0-1) of slope, distance to roads, inverse distance to building zones and elevation (Müller 2022). Select upper quartile of most marginal pixels and then select 90th percentile of pixels according to probability and increase probability to transition from Agriculture to non-agricultural classes. Simultaneously select pixels below upper quartile of

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-NAT	"Abandonment of poorly accessible marginal land"	NA	Alp_Past	Agri abandonment	Increase in probability of transition of marginal agricultural land to non-agricultural classes	Identify marginal agricultural land using an index (0-1) of slope, distance to roads, inverse distance to building zones and elevation (Müller 2022). Select upper quartile of most marginal pixels and then select 90th percentile of pixels according to probability and increase probability to transition from Agriculture to non-agricultural classes. Simultaneously select pixels below upper quartile of

Scenario	Scenario statement(Mayer et al., 2023)	Spatial zoning	LULC class target	Intervention ID	Mechanism overview	Mechanism detailed
EI-CUL	"Extensive cultivation of poorly accessible marginal land"	NA	Alp_Past	Agri maintenance	decrease in probability of marginal agricultural land transitioning to other classes	Identify marginal agricultural land using an index (0-1) of slope, distance to roads, inverse distance to building zones and elevation (Müller 2022). Select 90th percentile of most marginal pixels and then select 95th percentile of pixels according to probability and decrease probability to transition from Agriculture by 5%